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E-Mamba: Efficient Mamba network for hyperspectral and LiDAR joint classification

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ABSTRACT

Models based on Convolutional Neural Networks (CNNs) and Transformers have made significant advancements in the joint classification of hyperspectral image (HSI) and Light Detection and Ranging (LiDAR) data. However, these models also exhibit inherent limitations. CNNs, constrained by their network architecture, struggle to effectively capture long-range feature dependencies. On the other hand, while Transformers, with the self-attention mechanism, are adept at modeling global features, their scalability is hindered by the quadratic computational complexity. Fortunately, the recently emerged Mamba, a State Space Model (SSM), offers high computational efficiency while retaining the modeling capabilities of Transformers. In this paper, we present a preliminary attempt to apply the Mamba architecture to the joint classification task of HSI and LiDAR data. We utilize a dual-branch Siamese Visual State Space Block to extract distinctive features from the two heterogeneous modalities. The core module, 2D Selective Scan (SS2D), traverses paths in four different directions to generate multiple image patch sequences, with each sequence processed by individual S6 blocks. These sequences are then integrated and merged to capture the global features of the image. Considering the importance of inter-modal complementarity in improving classification performance, we introduce two lightweight Mamba feature extraction and fusion modules. The Cross-Mamba Feature Extraction Block is designed to enhance cross-modal information flow through features interaction. To better select relevant information from each modality, the Modality-Guided Mamba Fusion Block employs an attention mechanism to highlight pertinent information, further enhancing the model's discriminative ability. The aim is to reduce the number of training parameters while maintaining competitive classification performances. E-Mamba represents a pioneering effort in applying the Mamba architecture to the joint classification of HSI and LiDAR data, providing a foundation for future networks in multi-source remote sensing (RS) image classification based on Mamba. We demonstrate the effectiveness of E-Mamba through extensive experiments on three real-world datasets, achieving overall classification accuracies of 93.81%, 99.06%, and 91.47% on the Houston2013, Trento, and MUUFL datasets, respectively, surpassing a range of latest CNN-based and Transformer-based models. The code will be released at <https://github.com/zhangyiyan001/E-Mamba>.

1. Introduction

With advancements in Remote Sensing (RS) imaging technologies, various types of data have become increasingly accessible [1,2]. These data offer unprecedented opportunities for comprehensive and effective monitoring of surface cover [3–5]. However, single-modal RS data often fail to capture the complex characteristics of land covers, limiting their effectiveness [6–8]. To address this limitation, integrating multiple remote sensing data sources for classification has garnered significant

attention in recent years. In particular, the joint classification of hyperspectral image (HSI) and Light Detection and Ranging (LiDAR) data has emerged as a prominent research focus [9–13]. HSIs consist of hundreds of spectral bands, each corresponding to specific spectral reflection characteristics, forming “spectral fingerprints” for various land covers, thereby facilitating accurate classification [14–21]. However, HSIs have certain limitations, including the absence of elevation information and susceptibility to environmental factors that can alter

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spectral reflection characteristics, complicating the classification process [22–24]. In contrast, LiDAR, an active remote sensing technology, precisely measures the 3D structural information of the Earth's surface by emitting laser pulses and receiving their return signals [25–27]. LiDAR data provide detailed terrain and elevation information and are less affected by atmospheric conditions. Consequently, LiDAR data effectively compensate for the limitations of HSIs, and the complementary information from both sources can significantly enhance the accuracy and robustness of surface cover classification. Currently, joint classification techniques using HSIs and LiDAR are widely applied in various fields, such as environmental monitoring [28,29], precision agriculture [30,31], and urban planning [32,33].

In recent years, the rapid development of deep learning has significantly advanced remote sensing image processing technology. Convolutional Neural Networks (CNNs), known for their powerful feature extraction capabilities, have demonstrated broad application potential in tasks such as medical image segmentation [34,35], object detection [36,37], and face recognition [38,39]. Similarly, CNN-based methods have been widely applied to the joint classification of HSI and LiDAR data [10,11,40–44]. Leveraging CNNs' backbone frameworks, researchers have developed numerous network architectures to extract features from HSI and LiDAR data for integrated classification. These architectures typically achieve effective multimodal data fusion by extracting spectral and spatial features from HSI and elevation features from LiDAR data. However, due to the inherent limitations of CNNs concerning local receptive fields, these methods excel at capturing local features and modeling data but struggle to capture global contextual features. This limitation restricts the model's ability to fully optimize its performance on a larger scale. Specifically, CNNs extract local features through convolution operations, but each convolutional kernel perceives only the information within its local scope. While this issue can be partially mitigated by increasing the number of network layers or enlarging the convolutional kernel size, these solutions often lead to a significant increase in computational complexity and still fall short of fully capturing the global features of the data. Therefore, although CNN-based joint classification methods have achieved notable success in improving classification accuracy and robustness, the effective integration of local and global features remains a critical challenge.

The emergence of Transformers offers an effective solution to the challenges of local feature extraction, achieving significant success in Computer Vision (CV) and enabling performance that matches or even surpasses that of Convolutional Neural Networks (CNNs) [45–49]. Many researchers have successfully applied the Transformer architecture to the joint classification of HSI and LiDAR data [12,50–55]. Transformers excel in fusing information from both data sources through their self-attention mechanism, thereby enhancing classification accuracy and model robustness in the context of HSI and LiDAR data. However, the high computational cost of Transformer models remains a significant challenge. The computational complexity of the self-attention mechanism is quadratic, meaning that as the length of the input data increases, the computational and memory demands grow dramatically, leading to longer training times and higher resource consumption.

As an effective alternative to the self-attention mechanism, the State Space Model (SSM), particularly its recent variant Mamba, has gained recognition as a powerful feature extraction tool with linear complexity [56,57,58]. While sharing the ability of Transformers to capture long-range dependencies, Mamba offers significantly greater computational efficiency and reduced memory usage. This has led to increased interest in Mamba-based models across computer vision and, more recently, in remote sensing applications [59–61]. Building upon this foundation, but introducing fundamentally distinct architectural and methodological innovations, we present E-Mamba, a novel multi-modal classification framework for HSI and LiDAR data. While [61] concurrently explored Mamba for this task in their work, E-Mamba distinguishes itself through a unique two-stage fusion process and a

Siamese Visual State Space Block design with 2D selective scanning. The proposed model captures long-range dependencies in multi-modal data, overcomes the limited receptive fields of CNNs, and mitigates the substantial computational overhead of the Transformer's self-attention mechanism, while also addressing the limitations of existing Mamba implementations in handling 2D spatial data and multi-modal fusion. The main contributions of this paper are as follows:

- (1) To maximize the potential of the Mamba, we incorporate a dual-branch Siamese VSSB to extract distinctive features from both HSI and LiDAR modalities. This approach not only addresses the limitations of CNNs in local feature extraction but also enables global feature extraction from multisource images with linear computational complexity.
- (2) To improve the fusion of heterogeneous modalities, we employ a cross-multiplication mechanism to enhance the features from each modality and apply the SSM to the concatenated feature embeddings, producing the final fusion results.
- (3) Extensive experiments on three benchmark datasets have demonstrated that the proposed model outperforms other comparative methods, including CNN-based and Transformer-based models. The source code for this work is publicly available to facilitate contributions to future research.

The rest of the paper is organized as follows. Section 2 presents a detailed description of the related research work including traditional-based, CNN-based, Transformer-based, and Mamba-based methods. Section 3 provides an overview of the proposed model and a detailed analysis of the three components. Section 4 introduces the extensive experiments and presents the in-depth analysis of the experimental results. Section 5 summarizes the paper and points out the prospects for further research.

2. Related works

In this section, we provide a comprehensive review of joint HSI and LiDAR classification methods, encompassing traditional, CNN-based methods, and Transformer-based techniques. Additionally, we introduce latest works focused on HSI classification based on Mamba structure, offering valuable inspirations that contributed to our proposed model.

2.1. Traditional methods

Traditional HSI and LiDAR classification methods can be broadly categorized into filtering strategies, low-rank models, and composite kernels. Filtering methods primarily focus on reducing redundant information while preserving key features from other regions to effectively extract contextual and spatial characteristics. Common methods include morphological profiles [62], attribute profiles [63], and extinction profiles [64]. These methods are conceptually straightforward and allow for rapid HSI and LiDAR fusion, resulting in accurate classification. However, they tend to encounter challenges such as the curse of dimensionality and the high computational cost required in subsequent classification steps. Low-rank models and composite kernels offer effective solutions to these limitations [65]. Low-rank models assume that the features in HSI and LiDAR data can be represented in a lower-dimensional space, thus reducing the dimensionality of HSI and simplifying the fusion process. In contrast, composite kernel methods [42] address spectral, spatial, and elevation features by employing multiple kernel functions that extract different features. Nevertheless, designing and selecting appropriate kernel functions require careful calibration.

2.2. CNNs-based methods

Benefiting from their powerful feature extraction capabilities, CNNs automatically perform HSI and LiDAR feature extraction through end-to-end networks. Wu et al. [66] proposed an end-to-end network that employed CNNs and incorporated an advanced cross-channel reconstruction module for the efficient fusion of diverse RS data sources. Wang et al. [67] introduced a multiscale spatial spectral cross-modal attention network, where the multiscale pyramid convolution is used to extract multi-scale features, and the effective feature recalibration module is employed to enhance features at each scale and suppress irrelevant information. To capture deeper and more abstract discriminative features, Hang et al. [68] proposed a coupled CNN framework that fuses heterogeneous data with two branches: one capturing spectral-spatial features from HSI and the other extracting elevation details from LiDAR. Some researchers observed that a single level of feature fusion yielded only marginal performance improvements, prompting further exploration into multi-layered feature fusion from shallow to deep layers. et al. (2023) presented a network combining global and local features (NCGLF), leveraging CNNs for high-frequency feature extraction and transformers for low-frequency information, showcasing synergy between these techniques. Li et al. [69] presented a unique dual-channel long short-term memory neural network paired with a three-level fusion strategy, enhancing the fusion of HSI and LiDAR for superior classification. Additionally, Zhao et al. [70] proposed a hierarchical random walk network (HRWN) that integrates a dual-tunnel CNN with random walk layers, using LiDAR data to enforce spatial consistency and address fragmentation in classification maps. However, CNN-based methods still face inherent limitations in their receptive fields, and their ability for global feature modeling requires further improvement.

2.3. Transformers-based methods

In recent years, Transformer models have gained significant popularity in the field of CV. Numerous Transformer-based models have emerged for multimodal remote sensing image classification. For example, Ding et al. [54] presented a global-local transformer network, optimizing the convolution operator's ability to characterize locally correlated features and the transformer's proficiency in learning long-range dependencies. However, vanilla vision transformers tend to overlook local semantic information. To address this issue, Zhao et al. [53] proposed the Fractional Fourier Image Transformer, designed to accelerate transformer architectures while effectively and efficiently capturing valuable contextual information. Roy et al. [52] developed a multi-modal fusion transformer network, utilizing a multi-head cross-patch attention mechanism to extract complementary features from HS and LiDAR data, thereby achieving better generalization. Yao et al. [55] adapted the conventional vision Transformer with minimal alterations, introducing a cross-modality attention module to harness pixel-level spatial correlations in RS scenes. Zhang et al. [12] introduced the Multimodal Transformer Network (MTNet), which utilizes specialized HS spectral and spatial Transformers alongside LiDAR spatial Transformers, followed by a redesigned Transformer encoder to explore shared characteristics, offering a balance between modality-specific and fused feature learning. However, Transformers bear the burden of secondary computational complexity, which greatly increases memory and time costs. Additionally, Yang et al. [71] presented a LiDAR-guided cross-attention fusion method, using LiDAR data to steer HS band selection and employing cross-attention to fuse multimodal features, effectively reducing redundancy while boosting classification precision. Ni et al. [72] proposed the Multiscale Head Selection Transformer, incorporating a multiscale selection mechanism within the Transformer framework to integrate HS and LiDAR features at varying scales, enhancing robustness and adaptability to complex scenes.

2.4. Mamba-based methods

Mamba inherits the linear scalability in sequence length from state space models while incorporating the modeling capabilities of Transformers [56,57,59,73]. Unlike Transformers, which rely on explicitly storing the entire context for reasoning, Mamba combines selective scanning (S6) and enjoys fast inference with linear scaling of sequence lengths. As natural language processing and CV continue to converge, extending the high-performance Mamba architecture to applications in CV presents a promising avenue. Vim [74], a Mamba-based architecture, operates directly on image patch sequences similar to Vision Transformer (ViT). This architecture converts input images into flattened 2D patches, processes them through a linear projection layer with positional embeddings to preserve spatial information, and ultimately produces classification results via a class token. To more effectively capture 2D spatial information, VMamba [73] incorporates a Cross-Scan Module (CSM) that utilizes bidirectional scanning along both horizontal and vertical axes. The CSM transforms the input image into sequences of patches along these axes and processes them in four directions using selective state space models. This bidirectional application allows each pixel to integrate information from all other pixels across four distinct directions. All sequences are reverted to their original 2D layout to reconstruct the image, which is then merged to produce a new representation (scan merge). Currently, Mamba-based models are also being applied to HSI classification. Yao et al. [60] introduced SpectralMamba, which features piece-wise sequential scanning and gated spatial-spectral merging to enhance sequential learning in the state domain and correct spectral data. Li et al. [75] developed HSI-Mamba, which simultaneously models long-range interactions across entire images while adaptively integrating spatial and spectral information. Wang et al. [76] proposed a spatial-spectral state space model for HSI classification, comprising two selective SSMs for feature extraction in different dimensions—one focusing on spatial features and the other on spectral features—along with a spatial-spectral mixture gate to optimize fusion, achieving competitive classification performances. Liao et al. [61] proposed a joint classification network based on Mamba, which first processes LiDAR data using the Gradient Joint Algorithm (GJA) to extract the edge contour data of land distribution, and then uses Mamba for feature extraction and classification. This method effectively integrates the precise spectral information of hyperspectral data and the rich elevation information of LiDAR, addressing the challenge of fusing multimodal heterogeneous features. Li et al. [77] introduced an adaptive feature alignment network, using a global-local Mamba encoder to model the context through a 2D selective scanning mechanism, while also incorporating local bias to enhance the spatial features of local objects. In addition, the method proposed the SSE adaptive alignment and fusion module, which adaptively adjusts the relative positions of multi-source features to optimize the fusion effect.

3. Preliminaries

3.1. State space models

State Space Models (SSMs) are a class of mathematical models used to analyze and predict the behavior of dynamic systems that evolve over time. In general, SSMs can be defined using three parameters: $(\mathbf{A}, \mathbf{B}, \mathbf{C})$. These models transform time-dependent inputs $x(t) \in \mathbb{R}$ into a set of outputs $y(t) \in \mathbb{R}$ using a hidden state $h(t) \in \mathbb{R}^N$. The calculation process can be represented as:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t), \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t), \end{aligned} \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ represents the state transition matrix. The $\mathbf{B} \in \mathbb{R}^{N \times 1}$ and $\mathbf{C} \in \mathbb{R}^{1 \times N}$ are projection matrices. The Structured Space State Model (S4) is inspired by continuous systems used for 1D sequences

and utilizes the sampling timescale parameter Δ to convert a continuous state space model into a discrete SSM. Thus, the mapping of a continuous signal input $x(t)$ to output $y(t)$ is converted to sequence-to-sequence mapping: $x_k \rightarrow y_k$, which facilitates the processing of discrete labeled inputs, such as images and text.

This process can be divided into two stages: the first stage is discretization, where the continuous parameters $(\mathbf{A}, \mathbf{B})$ are converted to discrete parameters $(\bar{\mathbf{A}}, \bar{\mathbf{B}})$ by sampling the time scale parameter Δ . This conversion is typically achieved using Zero-Order Hold (ZOH), which leads to the discrete equation:

$$\begin{aligned}\bar{\mathbf{A}} &= e^{\Delta \mathbf{A}}, \\ \bar{\mathbf{B}} &= (\Delta \mathbf{A})^{-1} (e^{\Delta \mathbf{A}} - \mathbf{I}) \cdot \Delta \mathbf{B}, \\ h_k &= \bar{\mathbf{A}} h_{k-1} + \bar{\mathbf{B}} x_k, \\ y_k &= \mathbf{C} h_{k-1}.\end{aligned}\quad (2)$$

Where $\bar{\mathbf{A}} \in \mathbb{R}^{N \times N}$, $\bar{\mathbf{B}} \in \mathbb{R}^{N \times 1}$ and $\mathbf{C} \in \mathbb{R}^{1 \times N}$. The second stage is computation, the model computes the output by global convolution:

$$\begin{aligned}\bar{\mathbf{K}} &= (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \dots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}), \\ y &= x * \bar{\mathbf{K}},\end{aligned}\quad (3)$$

where L is the length of the input sequence x , and $\bar{\mathbf{K}} \in \mathbb{R}^L$ is a structured convolutional kernel.

3.2. Selective state space models (S6)

Previous linear time-invariant state-space models lack context awareness, causing S4 to give equal attention to all tokens, regardless of their varying importance. In practice, the significance of tokens changes dynamically during the training process. Therefore, it is more effective to focus on important content and dynamically adjust the attention level to match the complexity of the input. To address this, Mamba integrates a selectivity mechanism into the state space model, resulting in the Selective State Space Model (S6). Specifically, to operate on a sequence of inputs with batch size B , length L and D channels, Mamba applies the SSM independently to each channel. In Mamba, the matrices $\mathbf{B}$, $\mathbf{C}$, and Δ in S4 are functions of the inputs, allowing the model to adaptively tailor its behavior to the inputs. The discretization process after incorporating the selectivity mechanism into the model is as follows,

$$\bar{\mathbf{B}} = s_B(x) \quad (4)$$

$$\bar{\mathbf{C}} = s_C(x) \quad (5)$$

$$\Delta = \tau_A (\text{Parameter} + s_A(x)), \quad (6)$$

where $\bar{\mathbf{B}} \in \mathbb{R}^{B \times L \times N}$, $\bar{\mathbf{C}} \in \mathbb{R}^{B \times L \times N}$ and $\Delta \in \mathbb{R}^{B \times L \times D}$. $s_B(x)$ and $s_C(x)$ are linear functions that project the input x into a N -dimensional space, while $s_A(x)$ projects the hidden state dimension D linearly into the desired dimension. Through the above computations, the parameters Δ , $\mathbf{B}$, $\mathbf{C}$ become input functions with length L , transforming the time-invariant model into a time-varying model, thus achieving selectivity. With this selection mechanism, Mamba is capable of effectively modeling the complex interactions present in long sequences.

4. Methodology

4.1. Siamese Visual State Space Block (SVSSB)

The Siamese Visual State Space Block (SVSSB) is structured into two parallel branches, where both data types are processed through a Siamese network comprised of VSSBs with shared weight parameters, allowing for the joint extraction of spatial and spectral features and reducing the computational parameters (see Fig. 1). However, within the domain of Natural Language Processing (NLP), the S6 mechanism

of the State Space Model (SSM) is more adept at handling sequential data common to NLP tasks, but proves less effective with image data possessing a two-dimensional spatial structure. Drawing inspiration from VMamba, we employ a cross-scanning strategy to bridge the gap between 1D sequential data and 2D imagery.

Given the original multisource HSI and LiDAR data, denoted as $X_{hsi} \in \mathbb{R}^{H \times W \times C}$ and $X_{lidar} \in \mathbb{R}^{H \times W \times 1}$, both captured over the same geographical region, where C , H , and W correspond to the spectral channels, height, and width, respectively. HSI and LiDAR data differ significantly in channel dimensions due to their distinct acquisition methods, necessitating a 1×1 convolution to align LiDAR's single-channel data with HSI's multi-channel feature space. The processed HSI and LiDAR data can be symbolized as:

$$X_m = \{x_m^{(i,j)} \mid i = 0, 1, \dots, H-1, j = 0, 1, \dots, W-1\} \quad m = \{hsi, lidar\}, \quad (7)$$

where m refers to the specific modality data (HSI or LiDAR), and $x_m^{(i,j)}$ represents the pixel information at the spatial position (i, j) in the land cover map. For both HSI and LiDAR data, image cubes centered around the ground truth pixels are constructed, denoted as:

$$X_m^{(i,j)} = \begin{pmatrix} x_m^{[i-(p/2), j-(p/2)]} & x_m^{[i-(p/2), j-(p/2)+1]} & \dots & x_m^{[i-(p/2), j+(p/2)]} \\ x_m^{[i-(p/2)+1, j-(p/2)]} & x_m^{[i-(p/2)+1, j-(p/2)+1]} & \dots & x_m^{[i-(p/2)+1, j+(p/2)]} \\ \vdots & \vdots & \ddots & \vdots \\ x_m^{[i+(p/2), j-(p/2)]} & x_m^{[i+(p/2), j-(p/2)+1]} & \dots & x_m^{[i+(p/2), j+(p/2)]} \end{pmatrix} \in \mathbb{R}^{C \times p \times p} \quad (8)$$

where $(p/2) \leq i \leq H - (p/2) + 1$ and $(p/2) \leq j \leq W - (p/2) + 1$. To effectively leverage the spatial structure of the 2D image, we adopt a Cross Scan Strategy (CSS) for each image cube. The CSS is designed to traverse the spatial domain and transform any non-causal visual image into a sequence of sequential patches using a four-directional scanning approach. Specifically, the image cubes are expanded into sequences along both rows and columns, followed by scanning in four distinct directions, which can be referred in Fig. 2. The process is formalized as follows (see the Eq. (9) given as in Box 1):

where $S_{(m, type1)}^{(i,j)}$ represents the scanning order from the top left to the bottom right. The notation $V \rightarrow H$ indicates a horizontal scan from left to right, where after reaching the last element of a row, the scan moves down one row vertically, starting again from the leftmost element. This process continues until the bottom right pixel is reached, completing a full scanning sequence. Conversely, $S_{(m, type3)}^{(i,j)}$ denotes the scanning order from the bottom right to the top left, but with a variation in the scanning direction. The $H \rightarrow V$ represents a vertical scan from top to bottom, where after reaching the last element of a column, the scan shifts right to the next column, starting from the topmost element. This continues until the bottom right pixel is reached, forming an alternative scanning sequence. The $S_{(m, type2)}^{(i,j)}$ and $S_{(m, type4)}^{(i,j)}$ follow the reverse order, respectively. Consequently, we take the $S_{(m, type1)}^{(i,j)}$ as an illustrative example, feeding it into the Mamba architecture,

$$\begin{aligned}\mathbf{h}_{(m, type1)}^{(i,j)}(t) &= \bar{\mathbf{A}} \mathbf{h}_{(m, type1)}^{(i,j)}(t-1) + \bar{\mathbf{B}} \mathbf{S}_{(m, type1)}^{(i,j)}(t) \\ \mathbf{y}_{(m, type1)}^{(i,j)}(t) &= \mathbf{C} \mathbf{h}_{(m, type1)}^{(i,j)}(t) + \mathbf{D} \mathbf{S}_{(m, type1)}^{(i,j)}(t).\end{aligned}\quad (10)$$

Next, the output sequences obtained from different scanning routes are fused to form the output map.

$$S_{(output)} = \text{Merge}\{S_{(m, type1)}^{(i,j)}, S_{(m, type2)}^{(i,j)}, S_{(m, type3)}^{(i,j)}, S_{(m, type4)}^{(i,j)}\}, \quad (11)$$

By employing complementary traversal paths, SS2D allows each pixel in the image to effectively assimilate information from all other pixels in varying directions, thereby enabling the construction of global receptive fields.

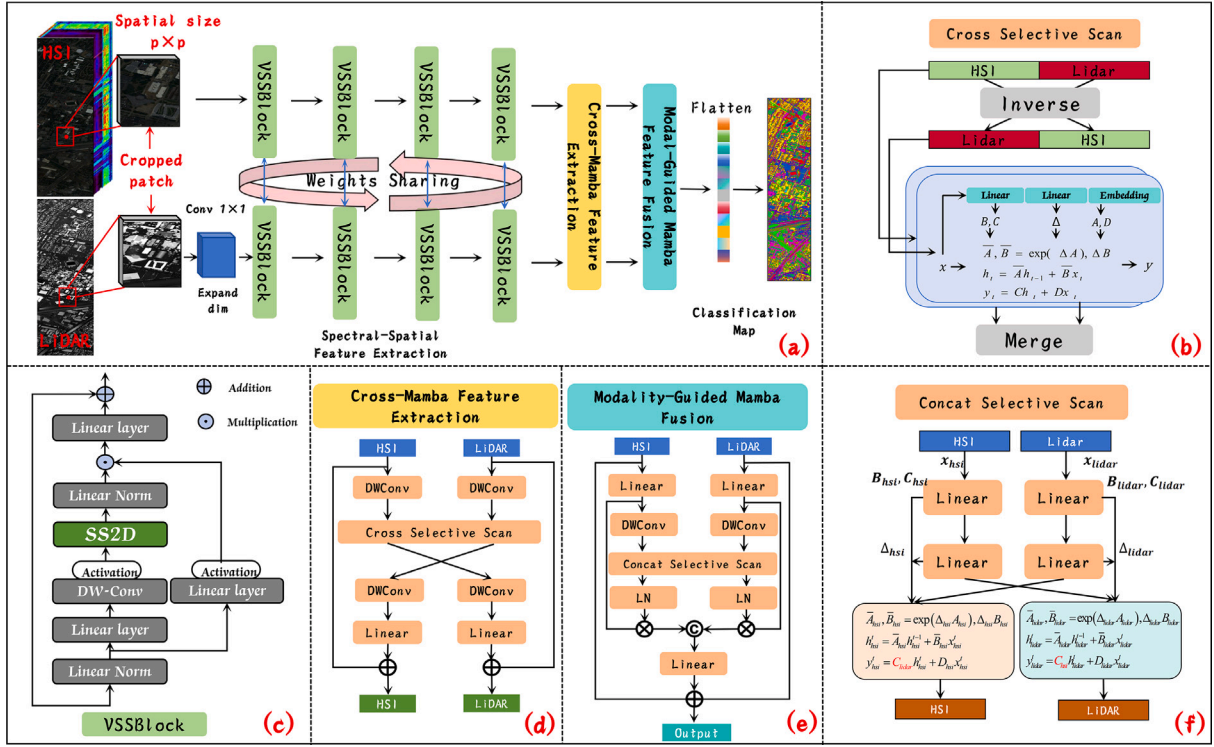


Fig. 1. The schematic diagram of the proposed E-Mamba and its components. (a) Framework of the E-Mamba. (b) Cross selective scan. (c) VSSBlock. (d) Cross-mamba feature extraction block. (e) Modality-Guided mamba fusion. (f) Concat selective scan.

$$\begin{aligned}
 S_{(m,type1)}^{(i,j)} &= [x_m^{[i-(p/2),j-(p/2)]}, x_m^{[i-(p/2),j-(p/2)+1]}, \dots, x_m^{[i+(p/2),j+(p/2)]}]^T & \text{top left} \rightarrow \text{bottom right} \quad (H \rightarrow V) \\
 S_{(m,type2)}^{(i,j)} &= [x_m^{[i+(p/2),j+(p/2)]}, x_m^{[i+(p/2),j+(p/2)-1]}, \dots, x_m^{[i-(p/2),j-(p/2)]}]^T & \text{bottom right} \rightarrow \text{top left} \quad (H \rightarrow V) \\
 S_{(m,type3)}^{(i,j)} &= [x_m^{[i-(p/2),j-(p/2)]}, x_m^{[i-(p/2)+1,j-(p/2)]}, \dots, x_m^{[i+(p/2),j+(p/2)]}]^T & \text{top left} \rightarrow \text{bottom right} \quad (V \rightarrow H) \\
 S_{(m,type4)}^{(i,j)} &= [x_m^{[i+(p/2),j+(p/2)]}, x_m^{[i+(p/2)-1,j+(p/2)]}, \dots, x_m^{[i-(p/2),j-(p/2)]}]^T & \text{bottom right} \rightarrow \text{top left} \quad (V \rightarrow H)
 \end{aligned} \tag{9}$$

Box I.

4.2. Cross-Mamba Feature Extraction Block (CMFEB)

Processed through the previously introduced SVSSB, the two parallel, weight-shared feature extraction branches perform feature extraction from shallow to deep layers for both HSI and LiDAR heterogeneous data, respectively. However, the features extracted from these two branches remain independent. For the task of joint classification of multi-source data, the effective fusion of these features is critical to improving classification performances. Hence, we propose the Cross-Mamba Feature Extraction Block (CMFEB), also built upon the Mamba architecture, which aims to fuse HSI and LiDAR feature embeddings at the feature-level.

The CMFEB takes two input feature sets and produces two outputs, maintaining the original shape of the input features. Initially, the two input features are processed through linear layers and Depthwise Convolutions (DWConv), after which they are passed to the Cross Selective Scan Module (CSSM), the core of the CMFEB. According to the selection mechanism inherent to Mamba, the system matrices B , C , and δ are dynamically generated based on the input, enabling the model's context-aware capabilities. Linear projection layers are employed to construct these matrices, where matrix C is used to decode information

from the hidden state h_k and produce the output y_k . Drawing inspiration from the cross-attention mechanism, which is widely utilized in multi-modal tasks, we seek to facilitate information exchange across multiple SSMS. To achieve this, we propose utilizing the C matrix generated by the complementary modality during the selective scan operation, allowing the SSM to reconstruct the output from the hidden state, guided by the other modality. Specifically, this process can be represented as:

$$\begin{aligned}
 \bar{A}_{hsi} &= \exp(\Delta_{hsi} A_{hsi}), \bar{A}_{lidar} = \exp(\Delta_{lidar} A_{lidar}) \\
 \bar{B}_{hsi} &= \Delta_{hsi} B_{hsi}, \bar{B}_{lidar} = \Delta_{lidar} B_{lidar}, \\
 h_{hsi}^{t-1} &= \bar{A}_{hsi} h_{hsi}^{t-1} + \bar{B}_{hsi} x_{hsi}^t, h_{lidar}^{t-1} = \bar{A}_{lidar} h_{lidar}^{t-1} + \bar{B}_{lidar} x_{lidar}^t, \\
 y_{hsi}^t &= C_{lidar} h_{hsi}^t + D_{hsi} x_{hsi}^t, y_{lidar}^t = C_{hsi} h_{lidar}^t + D_{lidar} x_{lidar}^t, \\
 y_{hsi} &= [y_{hsi}^1, y_{hsi}^2, \dots, y_{hsi}^T], y_{lidar} = [y_{lidar}^1, y_{lidar}^2, \dots, y_{lidar}^T],
 \end{aligned} \tag{12}$$

where x_m^t represents the input at time step t , and y_m^t denotes the output of the selective scan. The matrices C_{hsi} and C_{lidar} serve as the cross-modal matrices, employed to recover the outputs at each time step from the hidden states.

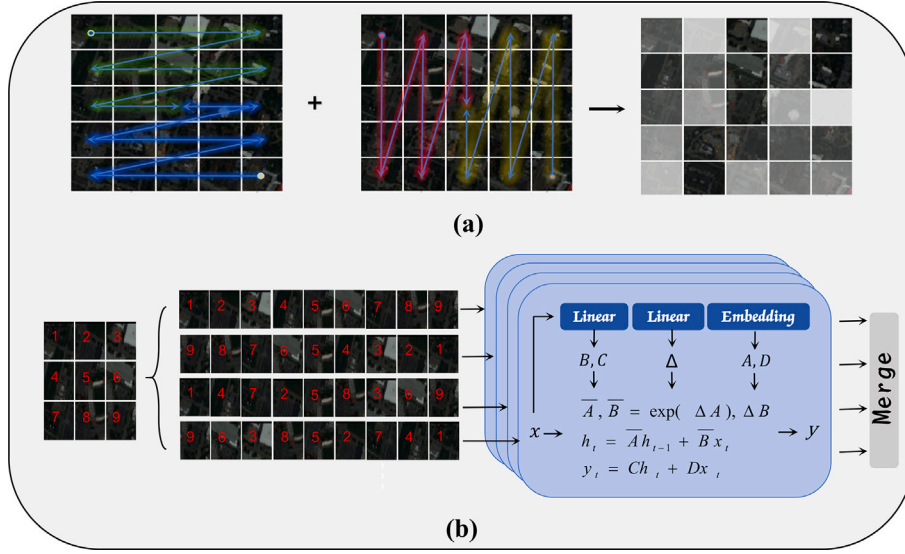


Fig. 2. (a) Cross-scan with four directions. (b) 2D-Selective-Scan operation.

4.3. Modality-guided Mamba Fusion Block (MMFB)

The features from the two modalities interact through the cross-selective scan operation, resulting in cross-modal-enhanced features in CMFEB. While the block accepts inputs from two heterogeneous data sources and outputs corresponding enhanced features, we propose the Modality-guided Mamba Fusion Block (MMFB) to further, integrating crucial information from both modalities. Previous Transformer-based approaches, constrained by quadratic scaling, typically divide inputs into small patches, limiting the ability to capture intra-token information. In contrast, leveraging Mamba's efficient long-sequence modeling capabilities, our MMFB directly processes concatenated features, thereby preserving maximal information from both modalities.

Specifically, the outputs $\hat{F}_{\text{hsi}}^k$ and $\hat{F}_{\text{lidar}}^k \in \mathbb{R}^{H_k \times W_k \times C_k}$ from CMFEB are first passed through linear and convolution layers, and subsequently fed into the Concat Selective Scan Module (CSSM). In the CSSM, the two features are flattened to $\mathbb{R}^{(H_k \times W_k) \times C_k}$ and concatenated along the sequence length dimension. This forms a sequence S_{Concat}^k with the shape $\mathbb{R}^{(2 \times H_k \times W_k) \times C_k}$. To further capture comprehensive information from both modalities, we apply an inverse scan to the concatenated sequence S_{Concat}^k , generating an additional sequence $S_{\text{Concat}}^k \in \mathbb{R}^{(2 \times H_k \times W_k) \times C_k}$. Each sequence is then processed by a 1D SSM, to capture long-range dependencies across both modalities, resulting in $\hat{S}_{\text{Concat}}^k$ and $\hat{S}_{\text{Inverse}}^k$. The inverse sequence is subsequently flipped back and combined with the processed concatenated sequence. The final summed sequence is then split to recover the two original input modalities. This process can be represented as:

$$\begin{aligned}
 \tilde{F}_{\text{hsi}}^k &= \text{DWConv}(\text{Linear}(\hat{F}_{\text{hsi}}^k)), \tilde{F}_{\text{lidar}}^k = \text{DWConv}(\text{Linear}(\hat{F}_{\text{lidar}}^k)), \\
 S_{\text{Concat}}^k &= \text{Concat}(\tilde{F}_{\text{hsi}}^k, \tilde{F}_{\text{lidar}}^k, \text{dim} = 0), \\
 S_{\text{Inverse}}^k &= \text{Inverse}(S_{\text{Concat}}^k), \\
 \hat{S}_{\text{Concat}}^k, \hat{S}_{\text{Inverse}}^k &= \text{SSM}(S_{\text{Concat}}^k), \text{SSM}(S_{\text{Inverse}}^k), \\
 \bar{F}_{\text{hsi}}^k, \bar{F}_{\text{lidar}}^k &= \text{Separate}(\hat{S}_{\text{Concat}}^k + \text{Inverse}(\hat{S}_{\text{Inverse}}^k)).
 \end{aligned} \tag{13}$$

After obtaining the scanned features $\tilde{F}_{\text{hsi}}^k$ and $\tilde{F}_{\text{lidar}}^k$, these are multiplied by two scaling parameters derived from $\hat{F}_{\text{hsi}}^k$ and $\hat{F}_{\text{lidar}}^k$, respectively. The resulting features are then concatenated along the channel dimension, producing a feature of shape $\mathbb{R}^{H_k \times W_k \times (2 \times C_k)}$. Finally, a linear projection layer is applied to reduce the feature shape to $\mathbb{R}^{H_k \times W_k \times C_k}$.

The visualizations of three different datasets including HSI pseudo-color and LiDAR grayscale maps are shown in Fig. 3.

5. Experimental results and analysis

5.1. Dataset description

(1) *Houston 2013 Dataset*: The Houston2013 dataset consists of HSI and LiDAR data with a spatial resolution of 2.5 m, encompassing an area of 349×1905 pixels. The HS data contain 144 spectral bands, covering wavelengths from 0.38 to 1.05 μm . This dataset includes 15 distinct land-cover categories.

(2) *Trento Dataset*: The Trento dataset comprises HSI and LiDAR data over an area of 166×600 pixels with a spatial resolution of 1 m. The HS data consist of 63 spectral bands, covering wavelengths from 0.42 to 0.99 μm . This dataset includes 6 distinct land-cover categories.

(3) *MUOFL Dataset*: The MUOFL Gulfport¹ dataset was collected over the Gulf Park campus of the University of Southern Mississippi, Long Beach, MS, USA, in November 2010. In this dataset, the HSI data comprises 64 spectral bands ranging from 0.38 to 1.05 μm , while the LiDAR data include 2 rasters with a wavelength of 1.06 μm . The dataset covers an area of 325×220 pixels and features 11 different land-cover classes.

5.2. Experimental configuration

- *Evaluation Metrics*: The quantitative classification performance is assessed using three metrics: Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ), and accuracy for each class. These indices range from 0 to 1, where higher values signify superior performance. OA, AA, and κ are specifically derived by aggregating all pixel-level predictions and comparing them to the corresponding ground truth labels

$$\begin{aligned}
 OA &= \frac{N_c}{N_a}, \\
 AA &= \frac{1}{K} \sum_{j=1}^K \frac{N_c^j}{N_a^j}, \\
 \kappa &= \frac{OA - P_e}{1 - P_e}.
 \end{aligned} \tag{14}$$

¹ <https://github.com/GatorSense/MUOFLGulfport>.

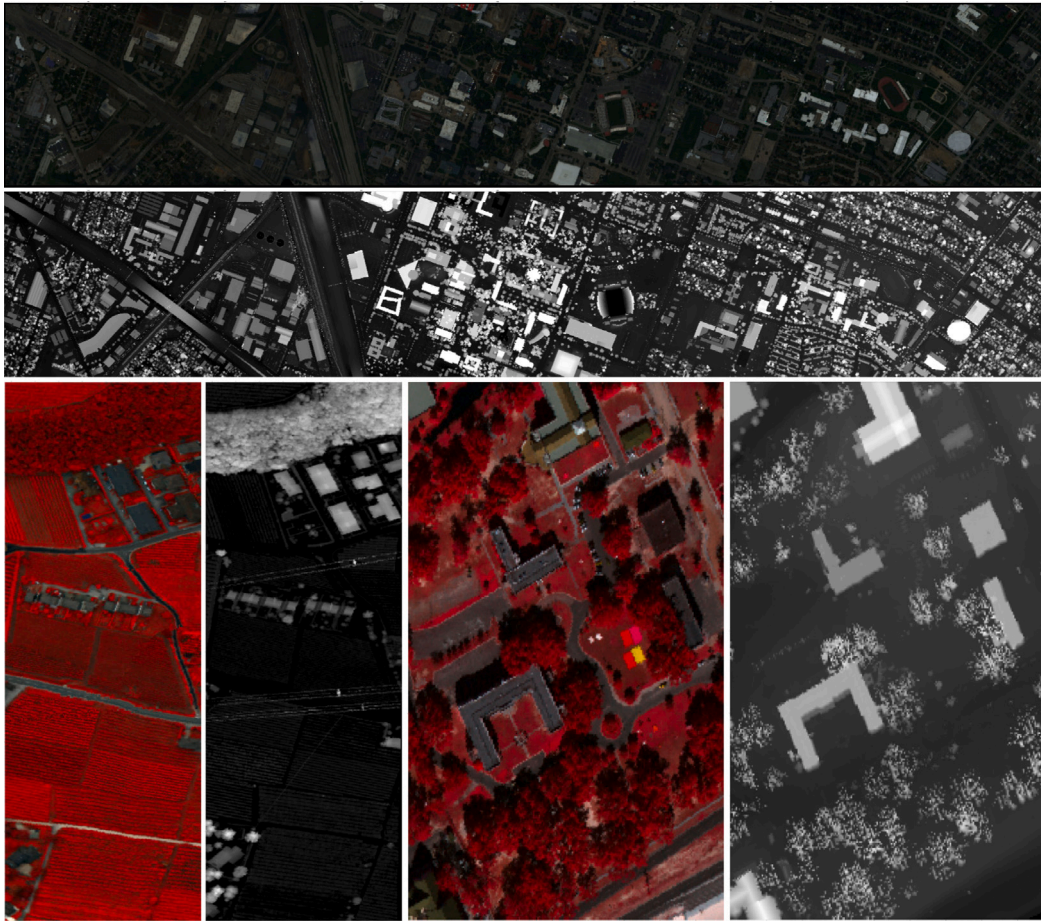


Fig. 3. Visualization of the HSI pseudo-color maps and LiDAR grayscale maps for the three datasets. The Houston dataset at the top, Trento dataset at the bottom left and MUUFL dataset at the bottom right.

Here N_c and N_a represent the number of correctly classified samples and the total number of test samples, while N_c^j and N_a^j correspond to these values for a specific j th class. The calculation of the κ addresses class imbalance by incorporating the hypothetical probability of chance agreement P_e , as defined by

$$P_e = \frac{\sum_{j=1}^K N_a^j \times N_p^j}{(N_a)^2}. \quad (15)$$

- **Implementation details:** The experiments were executed on a Linux PC server equipped with 256 GB of RAM and an RTX 3090 GPU. The proposed model was implemented using PyTorch and optimized with the AdamW optimizer. A total of 300 epochs were conducted. The initial learning rates were set to 0.0003 for the Houston dataset, 0.0001 for the Trento dataset, and 0.001 for the MUUFL dataset, with a decay rate of 0.0001. Additionally, the learning rate was halved every 30 epochs.
- **Comparison with other models:** Various CNN-based and transformer-based models were selected for comparative experiments, with their detailed configurations outlined below.

1. The CALC² [78] is an adversarial feature learning framework comprising both encoding and decoding components. The encoding section consists of a 2D convolution with a 3×3 kernel size, followed by batch normalization, ReLU

activation, and max pooling. The pooling window size is 2×2 , and the number of convolution kernels is set to 32, 64, and 128. The decoding section includes upsampling, a 2D convolution with a 3×3 kernel, batch normalization, and ReLU activation.

2. The GAMF³ [79] considers the correlation and complementarity between multi-sensor data via parameter sharing, employing Gaussian tokenization for feature transformation. The fusion module utilizes modality-specific tokens to construct an undirected weighted graph, and two fully connected layers classify the fused embeddings to produce the final results.
3. The GLTNet⁴ [54] is characterized by its multi-scale aggregated convolutional encoder and global spectral feature learning module, which combines CNNs and Transformer components to enable detailed feature extraction across both global and local scales.
4. The M2FNet⁵ [80] utilizes a multiscale 3D–2D mixed CNN architecture to extract spatial–spectral features from HSI data and elevation features from LiDAR data. These extracted features are processed by an innovative encoder, which includes a feature enhancement module that integrates mathematical morphology with a dilated

² <https://github.com/Ding-Kexin/CALC>.

³ <https://github.com/tyust-dayu/GAMF>.

⁴ <https://github.com/Ding-Kexin/GLT-Net>.

⁵ <https://github.com/cupid6868/M2FNet.git>.

convolutional module inspired by the self-attention mechanism of standard Transformer encoders.

5. MS2CANet⁶ [67] leverages pyramid convolution to capture multiscale features from multisource remote sensing imagery. An effective feature recalibration module is incorporated to amplify relevant features and suppress extraneous information across varying scales.
6. The S²ENet⁷ [81] model integrates two feature enhancement modules: the Spatial Attention Enhancement Module and the Spectral Attention Enhancement Module. The former enhances the spatial representations of LiDAR data by leveraging HSI features, while the latter refines the spectral representations of HSI data using LiDAR information. The results were reproduced following the hyperparameter configurations specified in the original study.
7. The MFT⁸ [52] employs Conv3D and HetConv2D layers in sequence to process raw HSI data, while LiDAR data undergoes pixel or channel tokenization. Feature integration is achieved through cross-attention between HSI and LiDAR patches.
8. The ExViT⁹ [55] incorporates three blocks that combine separable convolutions and transformer encoders, enabling independent processing of HSI and LiDAR data before integrating them for feature synthesis.
9. The HLMamba¹⁰ [61] first performs a gradient joint algorithm on LiDAR data to extract edge contour information. A multimodal feature extraction module (MFEM), based on CNNs, then extracts features from HSI and LiDAR data.

5.3. Parameter experiments

(1) *Impact of image patch size*: The analysis of the proposed method's classification performance under varying image patch sizes underscores the significance of selecting an optimal spatial size for effective feature extraction and accurate classification. Fig. 4 presents the overall accuracy (OA), average accuracy (AA), and κ values across the Houston, Trento, and MUUFL datasets, with spatial sizes ranging from 5×5 to 15×15 . For the Houston dataset, a clear improvement in classification metrics is observed as the spatial size increases from 5×5 to 9×9 , with all three metrics—OA, AA, and κ —peaking at 9×9 . This indicates that a moderate spatial size is most effective in capturing relevant features while minimizing the inclusion of irrelevant information. However, beyond 9×9 , a slight decline in performance occurs, likely due to the inclusion of excessive surrounding pixels that do not contribute meaningfully to the classification task. In the case of the Trento dataset, the performance metrics remain consistently high from 5×5 to 9×9 , with a slight peak at 9×9 . However, a noticeable decline in all three metrics is evident when the spatial size exceeds 11×11 , particularly at 15×15 . This sharp decrease can be attributed to the dataset's characteristics, where larger spatial sizes may introduce noise and redundant information, thereby impairing the model's ability to accurately classify targets. For the MUUFL dataset, optimal performance is achieved at smaller spatial sizes, specifically at 5×5 . As the patch size increases, a significant drop in OA, AA, and κ is observed, with the lowest performance recorded at 15×15 . This suggests that smaller patches are more effective for the MUUFL dataset, likely due to its unique feature distribution, which benefits from less contextual but more focused information. Based on these observations, the optimal patch size for achieving high classification accuracy while maintaining

computational efficiency is 9×9 for the Houston and Trento datasets, and 7×7 for the MUUFL dataset.

(2) *Impact of training ratio*: We conducted experiments by varying the training sample ratios (0.1, 0.3, 0.5, 0.7, 0.9) for the Houston, Trento, and MUUFL datasets. The overall accuracies (OA) of several models, including CALC, GAMF, GLTNet, M2FNet, MS2CANet, S²ENet, MFT, ExViT, and the proposed method, were evaluated. The results are presented in Fig. 5, illustrating the relationship between training ratio and model performance. For the Houston dataset, the proposed method exhibits a consistent upward trend in OA as the training ratio increases. Notably, even with only 0.1 of the training samples, the proposed method outperforms some models that use larger training ratios. As the training ratio reaches 0.3 and beyond, the proposed method continues to surpass other models, demonstrating its robustness and efficiency in learning from limited data. While models like CALC and GLTNet also show strong performance, they plateau at higher training ratios, indicating a less effective use of additional training data compared to the proposed method. In the Trento dataset, the proposed method once again leads in performance across different training ratios, particularly at lower ratios (0.1 to 0.3), where it outperforms all other models. As the training ratio increases, the OA of the proposed method continues to rise, whereas models such as MS2CANet and S²ENet exhibit relatively flat or even declining trends, potentially due to overfitting or reduced generalization capability with larger training sets. For the MUUFL dataset, the proposed method maintains strong performance across all training ratios, consistently ranking among the top models. The improvement in OA is more pronounced as the training ratio increases from 0.1 to 0.5, after which the growth rate stabilizes. In contrast, models like ExViT and S²ENet show slower improvements and eventually lag behind the proposed method, especially at higher training ratios. This emphasizes the proposed method's ability to effectively utilize available data for enhanced classification accuracy. Overall, the results demonstrate that the proposed method consistently achieves superior classification performance across varying training sample sizes, excelling particularly in scenarios with limited training data. Its robustness and scalability with larger datasets further underscore its potential for broader application in HSI and LiDAR data classification tasks.

(3) *The impact of layers of VSSBlock*: To examine the effect of VSSBlock depth on the proposed method's classification performance, we conducted experiments varying the depth from 1 to 4 layers. The results are illustrated in Fig. 6. As the depth of the VSSBlock increases from 1 to 4 layers, there is a consistent improvement in classification metrics, including OA, AA, and κ . The best performance is observed at a depth of 4 layers across all three datasets, suggesting that increasing the VSSBlock depth enables the model to capture more complex features, thereby improving classification accuracy. Consequently, a VSSBlock depth of 4 layers offers optimal classification performance for these datasets, providing better feature representation and higher accuracy compared to shallower configurations.

(4) *Computational Complexity Analysis*: To demonstrate the lower computational complexity of our method, we compared the OA, AA, and the number of computational parameters on the Trento and MUUFL datasets against other methods. The results, displayed in Fig. 8, show that our proposed method achieved the best performance in both OA and AA while utilizing fewer parameters. Comparative methods such as S²ENet and MS2CANet, although delivering strong classification accuracy, require significantly more parameters than E-Mamba, which poses challenges to the timeliness of HSI and LiDAR classification tasks. Moreover, the figure clearly highlights that the bubble representing GAMF has the largest area, indicating its massive number of parameters, despite its unsatisfactory classification performance on these two datasets. Other methods, such as M2FNet, have fewer parameters but suffer from significantly lower classification accuracy compared to our method. We also conducted a new experiment to compare the training

⁶ <https://github.com/junhengzhu/MS2CANet>.

⁷ <https://github.com/likyoo/Multimodal-Remote-Sensing-Toolkit>.

⁸ <https://github.com/AnkurDeria/MFT>.

⁹ <https://github.com/jingyao16/ExViT>.

¹⁰ <https://github.com/Dilingliao/HLMamba>.

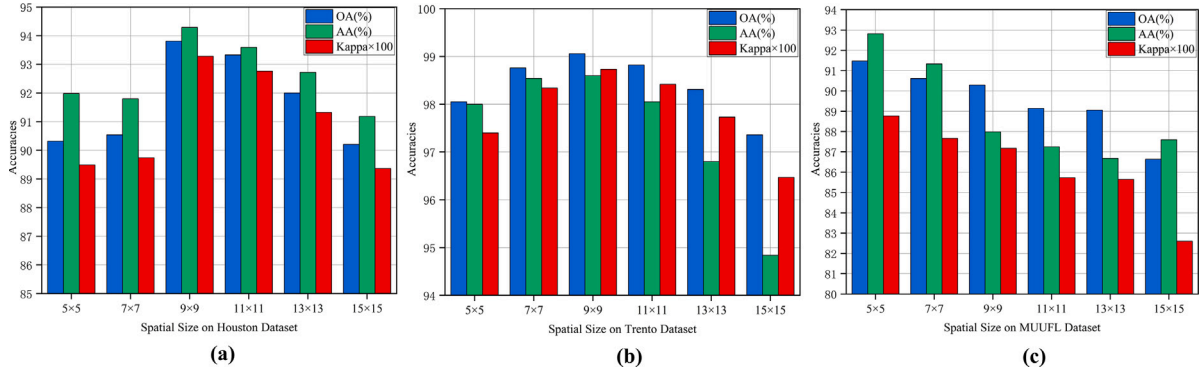


Fig. 4. Impact of the spatial size of the image cube on three datasets from 5×5 to 15×15 . (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

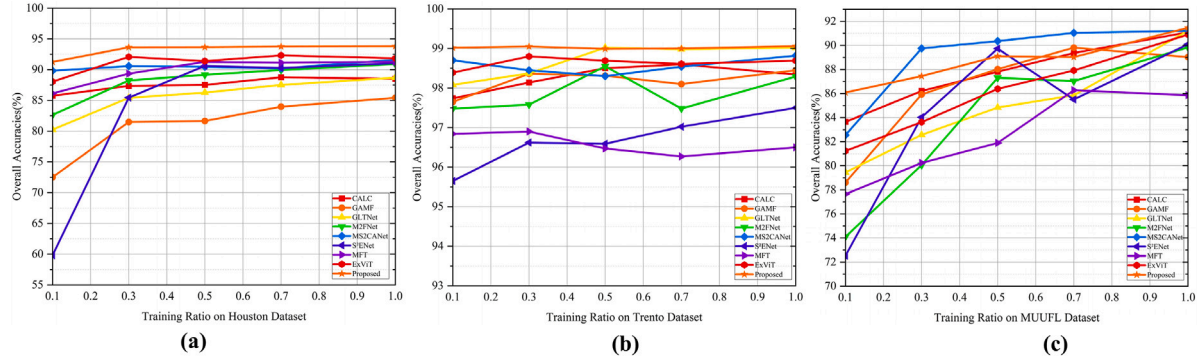


Fig. 5. Impact of different training ratios on three datasets from 0.1 to 0.9 at 0.2 intervals. (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

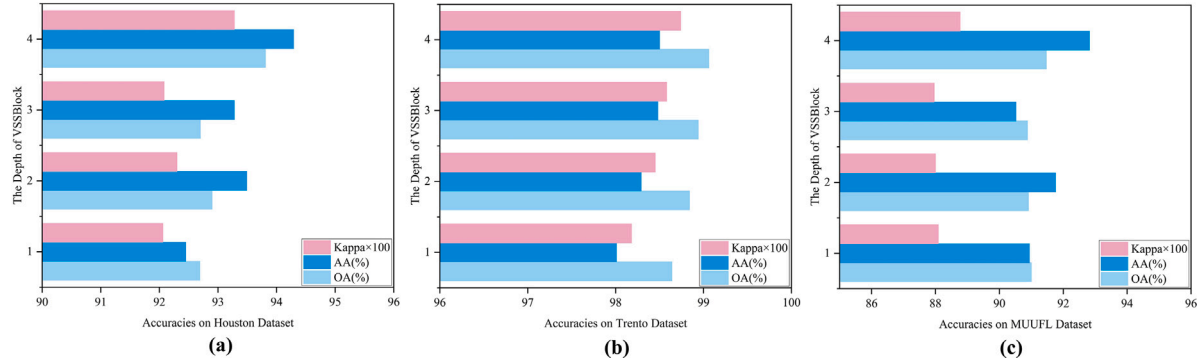


Fig. 6. Impact of the depth of VSSBlock on three datasets from 1 to 4 layers. (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

times of different methods across multiple datasets shown in Table 6. Overall, our method not only achieves superior classification performance with fewer parameters but also maintains efficient training times, reinforcing its practicality and effectiveness for HSI and LiDAR classification tasks.

5.4. Comparison with other models

Based on the quantitative experimental results presented in the tables for the Houston, Trento, and MUUFL datasets, the proposed method consistently demonstrates superior classification performance compared to other models. The quantitative classification results of the various comparison methods across the three datasets are reported in Tables 1–3. For the Houston dataset, the proposed method achieves the highest OA of 93.81%, surpassing all other methods, with notable performance in classes such as “Trees”, “Railway”, and “Running track”,

where it either matches or exceeds the accuracy of competing models. The AA and κ scores further underscore the method’s robustness, with values of 94.29% and 93.28%, respectively, indicating strong generalization and consistent classification across different land cover types. In the case of the Trento dataset, the proposed method once again outperforms the competing models, achieving the highest OA of 99.06%. The method particularly excels in the “Apple” and “Woods” classes, where it demonstrates perfect or near-perfect classification accuracy. The AA of 98.6% and κ of 98.74% further emphasize its effectiveness in capturing the complex spectral and spatial features inherent in the Trento dataset. For the MUUFL dataset, the proposed method maintains its leading performance with the highest OA of 91.47%, showcasing its strength in classes such as “Trees”, “Building”, and “Cloth panels”. The method’s AA of 90.45% and κ of 88.77% highlight its reliable performance across diverse urban and natural land covers, consistently outperforming other comparative models. The

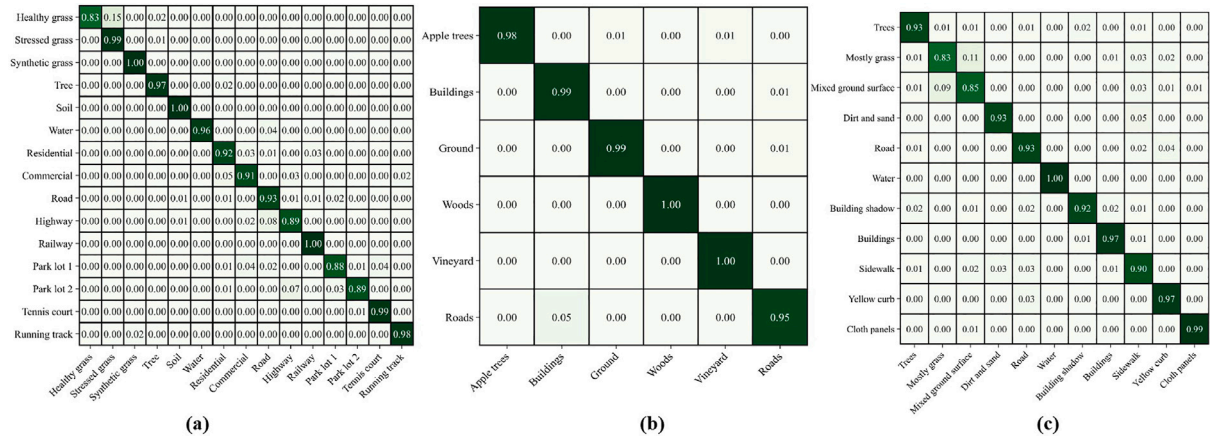


Fig. 7. The confusion matrices obtained by the proposed method on three datasets. (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

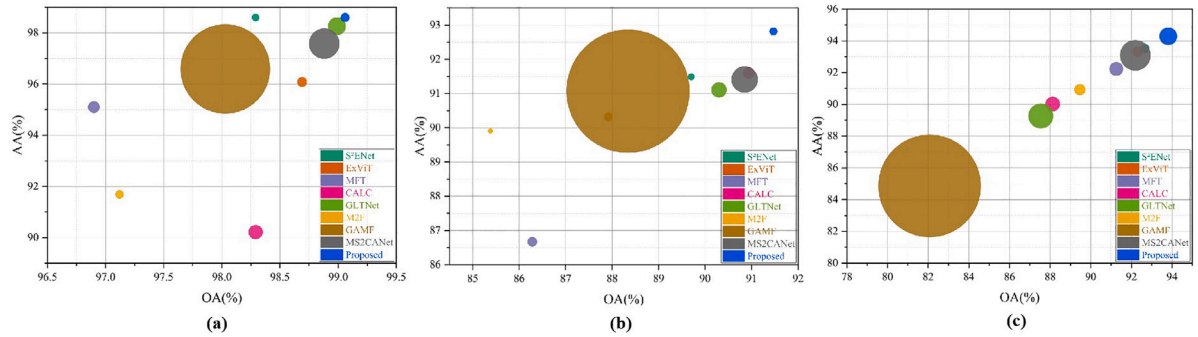


Fig. 8. Comparison of different methods in terms of OA, AA and training parameter on different dataset. The X-axis represents OA values and the Y-axis shows AA values. The size of bubble reflect the number of training parameters for each model. (a) Trento dataset. (b) MUUFL dataset.

Table 1

Classification results of different methods on Houston dataset. The best results are in bold and the second best results are underlined.

Class no.	Training/Testing	CALC	GAMF	GLTNet	M2FNet	MS2CANet	S ² E-Net	MFT	ExViT	HLMamba	Proposed
1	Health grass (198/1053)	80.91	83.10	83.10	92.12	82.43	83.10	82.11	83.74	82.18	82.81
2	Stressed grass (190/1064)	83.65	84.68	84.02	84.59	85.15	99.44	93.44	<u>99.46</u>	98.72	99.06
3	Synthetic grass (192/505)	91.88	100.00	96.24	98.22	99.40	100.00	<u>99.43</u>	96.63	98.06	99.80
4	Trees (188/1056)	90.34	87.97	92.04	97.82	<u>99.81</u>	100.00	97.02	99.10	99.93	97.35
5	Soil (186/1056)	100.00	99.62	100.00	96.02	<u>99.91</u>	100.00	99.35	98.47	100.00	100.00
6	Water (182/143)	100.00	93.01	<u>97.90</u>	95.80	100.00	95.8	96.3	96.97	98.15	95.80
7	Residual (196/1072)	86.38	83.58	<u>87.59</u>	88.15	89.37	<u>92.44</u>	95.6	88.33	91.58	<u>92.44</u>
8	Commercial (191/1053)	81.48	76.16	78.82	90.22	89.55	96.68	87.70	90.19	90.32	<u>90.69</u>
9	Road (193/1059)	81.49	69.97	92.26	<u>92.35</u>	85.08	86.69	91.47	87.82	92.85	93.39
10	Highway (191/1036)	68.53	53.19	68.44	52.99	87.64	81.47	73.48	<u>83.36</u>	88.18	89.48
11	Railway (181/1054)	100.00	74.86	82.73	91.08	97.91	93.26	94.69	95.85	98.52	<u>99.91</u>
12	Parking lot 1 (192/1041)	96.45	85.21	94.52	<u>97.11</u>	98.17	85.4	88.03	87.99	85.77	87.99
13	Parking lot 2 (184/285)	90.18	83.51	91.58	87.72	90.88	88.42	84.87	<u>92.38</u>	93.28	88.77
14	Tennis court (181/247)	99.19	100.00	95.55	100.00	92.71	100.00	100.00	<u>99.34</u>	98.25	99.19
15	Running track (187/473)	<u>99.79</u>	98.10	94.29	<u>99.79</u>	98.31	100.00	100.00	100	97.05	97.67
OA (%)		88.12	82.07	87.54	89.46	92.19	<u>92.66</u>	91.25	92.31	92.97	93.81
AA (%)		90.02	84.86	89.27	90.93	93.09	<u>93.51</u>	92.23	93.31	94.19	94.29
$\kappa \times 100$ (%)		87.12	80.56	86.54	88.58	91.58	<u>92.03</u>	90.51	91.67	93.10	93.28

confusion matrices obtained by the proposed model can be seen in Fig. 7.

5.5. Visual evaluation

Based on the classification maps for the Houston, Trento, and MUUFL datasets across different methods, several key observations can be made regarding the performance of each method in accurately capturing the ground truth labels. The corresponding classification maps obtained by various methods in three datasets are shown in Figs. 9–11. For the Houston dataset, the proposed method demonstrates

superior classification performance, with a more refined and accurate representation of the various land cover types compared to other methods. The proposed method shows clear boundaries between different classes, particularly in complex areas where other methods like CALC and M2FNet tend to blur the distinctions between similar classes, such as different types of grass and road surfaces. In the Trento dataset, the proposed method again excels, capturing the distinct class boundaries with high precision, especially in areas where buildings and vineyards are adjacent. Compared to other methods like MS2CANet and ExViT, the proposed method minimizes misclassification and better preserves the structure of smaller classes, such as roads and trees, which are

Table 2

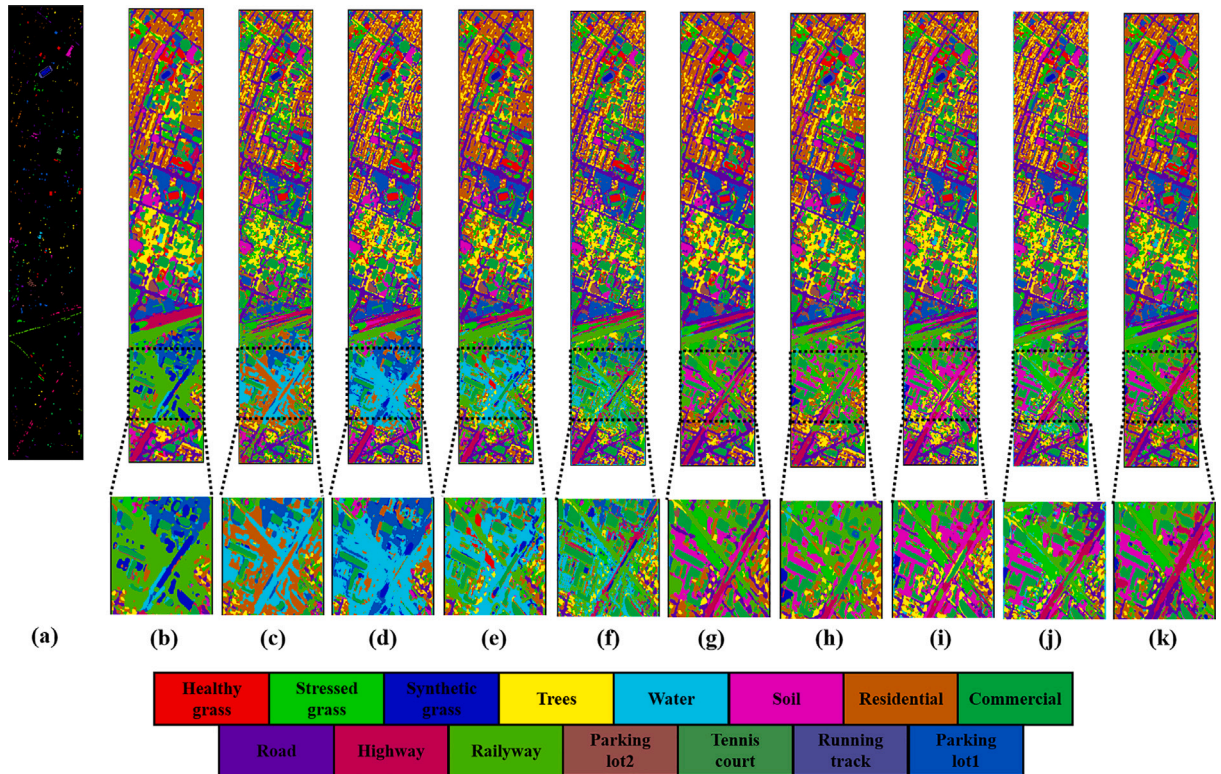
Classification results of different methods on Trento dataset. The best results are in bold and the second best results are underlined.

Class no.	Training/Testing	CALC	GAMF	GLTNet	M2FNet	MS2CANet	S ² ENet	MFT	ExViT	HLMamba	Proposed
1	Apple (129/3905)	99.95	98.39	99.00	<u>99.36</u>	99.08	97.08	91.92	98.04	99.08	98.21
2	Buildings (125/1064)	99.64	99.82	99.28	99.82	96.51	100.00	<u>99.97</u>	99.75	99.06	99.14
3	Ground (1058/374)	50.80	93.58	<u>98.93</u>	74.60	95.45	94.39	97.54	86.18	98.03	99.47
4	Woods (154/8969)	99.83	98.75	100.00	99.99	100.00	100.00	100.00	99.78	100.00	100.00
5	Vineyard (184/10317)	99.71	98.68	100.00	100.00	100.00	99.88	<u>99.96</u>	100.00	99.14	99.71
6	Roads (122/3525)	91.42	92.14	92.37	76.38	<u>94.43</u>	88.37	81.23	92.71	92.17	95.05
OA (%)		98.29	98.03	<u>98.99</u>	97.12	98.88	98.29	96.9	98.69	98.66	99.06
AA (%)		90.22	96.89	<u>98.26</u>	91.69	97.58	96.62	95.10	96.08	98.06	98.60
$\kappa \times 100$ (%)		97.71	97.36	<u>98.65</u>	96.14	98.54	97.71	95.87	98.25	98.06	98.74

Table 3

Classification results of different methods on MUUFL dataset. The best results are in bold and the second best results are underlined.

Class no.	Training/Testing	CALC	GAMF	GLTNet	M2FNet	MS2CANet	S ² ENet	MFT	ExViT	HLMamba	Proposed
1	Trees (150/23246)	<u>93.09</u>	89.28	91.31	84.29	94.63	90.66	90.24	88.05	92.32	92.98
2	Mostly grass (150/4270)	85.95	89.95	78.71	83.88	85.49	<u>86.31</u>	74.07	81.90	86.58	83.13
3	Mixed ground surface (150/6882)	81.24	73.57	86.26	80.24	75.83	79.69	76.30	77.70	85.98	84.55
4	Dirt and sand (150/1826)	97.29	94.39	<u>97.08</u>	91.71	93.68	93.02	93.61	92.5	92.61	93.44
5	Road (150/6887)	<u>91.72</u>	89.49	90.73	86.43	90.10	90.42	80.96	90.14	89.70	92.61
6	Water (150/466)	100.00	97.47	99.37	100.00	97.78	99.68	<u>99.72</u>	99.72	98.57	99.68
7	Building shadow (150/2233)	95.10	94.77	94.19	97.31	93.57	93.76	91.02	<u>95.35</u>	93.78	91.74
8	Building (150/6240)	94.88	96.19	<u>96.37</u>	87.80	95.80	95.52	95.78	96.19	96.87	96.93
9	Sidewalk (150/1385)	78.30	78.38	81.13	87.29	88.58	90.28	66.25	85.31	87.98	<u>89.80</u>
10	Yellow curb (150/183)	90.91	100.00	87.88	90.91	90.91	87.88	88.46	89.74	92.82	96.97
11	Cloth panels (150/269)	99.16	<u>98.32</u>	99.16	99.16	99.16	99.16	96.95	96.95	98.37	99.16
OA (%)		<u>90.94</u>	88.34	90.30	85.38	90.85	89.70	86.28	87.92	91.01	91.47
AA (%)		<u>91.60</u>	91.07	91.11	89.91	91.41	91.49	86.67	90.32	92.32	92.82
$\kappa \times 100$ (%)		<u>88.08</u>	84.78	87.28	81.22	88.05	86.53	82.17	84.36	87.62	88.77

**Fig. 9.** Classification maps of different models on the Houston dataset. (a) Ground Truth. (b) CALC. (c) GAMF. (d) GLTNet. (e) M2FNet. (f) MS2CANet. (g) S²ENet. (h) MFT. (i) ExViT. (j) HLMamba. (k) Proposed.

often difficult to distinguish. For the MUUFL dataset, the proposed method shows a clear advantage in accurately classifying mixed ground surfaces and building areas. It manages to reduce the noise present in the classification outputs of methods like S²ENet and GLT, leading to a more coherent and reliable classification map. The proposed method

also performs better in consistently identifying smaller classes like dirt and sand, and building shadows, which are often misclassified by other approaches.

Overall, the proposed model consistently outperforms other techniques across all three datasets, particularly in complex urban and

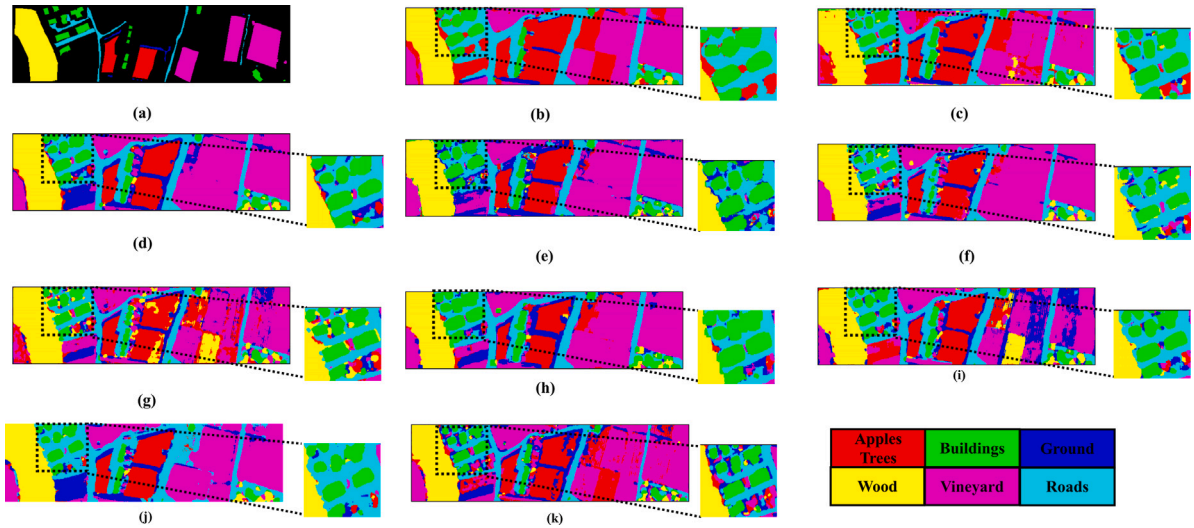


Fig. 10. Classification maps of different models on the Trento dataset. (a) ground truth. (b) CALC. (c) GAMF. (d) GLTNet. (e) M2FNet. (f) MS2CANet. (g) S²ENet. (h) MFT. (i) ExViT. (j) HLMamba. (k) proposed.

Table 4

Ablation studies results in terms of OA, AA, κ regarding data source usage.

Metrics	Houston			Trento			MUUFL		
	OA (%)	AA (%)	$\kappa \times 100$ (%)	OA (%)	AA (%)	$\kappa \times 100$ (%)	OA (%)	AA (%)	$\kappa \times 100$ (%)
Only HSI	91.87	92.32	91.17	96.76	94.88	95.56	89.11	89.88	94.31
Only LiDAR	53.86	52.74	49.99	87.01	77.91	82.03	74.78	75.40	67.53
HSI+LiDAR	93.81	94.29	93.28	99.06	98.6	98.74	91.47	92.82	88.77

Table 5

Ablation study results on different combinations of MMFB and CMFEB in Terms of OA, AA, and κ on three datasets.

MMFB	CMFEB	Houston			Trento			MUUFL		
		OA (%)	AA (%)	$\kappa \times 100$ (%)	OA (%)	AA (%)	$\kappa \times 100$ (%)	OA (%)	AA (%)	$\kappa \times 100$ (%)
	✓	92.86	93.43	91.87	98.96	98.65	98.83	90.88	92.44	88.24
✓		92.51	93.65	92.41	98.72	98.78	98.88	90.83	92.46	88.51
✓	✓	93.81	94.29	93.28	99.06	98.60	98.74	91.47	92.82	88.77

Table 6

Comparison of different methods across datasets in terms of training time(s).

Datasets	CALC	GAMF	GLTNet	M2FNet	MS2CANet	S ² ENet	MFT	ExViT	HLMamba	Proposed
Houston	362.5	1028.8	651.9	458.8	310.8	299.7	358.6	362.5	648.6	318.6
Trento	265.8	815.9	587.3	358.7	287.6	248.3	312.6	352.5	504.1	304.6
MUUFL	292.5	906.8	622	452.6	318.5	287.6	380.6	370.3	562.9	315.6

heterogeneous environments. It effectively captures detailed spatial and spectral features, leading to more accurate and visually consistent classification maps that align closely with the ground truth. This highlights the robustness and effectiveness of the proposed method in handling diverse and challenging remote sensing data.

5.6. Ablation studies

We conducted ablation studies to evaluate the individual contributions of the two heterogeneous data sources, HS and LiDAR, to the classification task (see Table 4). Based on the provided results, for the Houston dataset, the fused multisource data achieves an OA of 93.81%, significantly outperforming the 91.87% achieved using only HSI and the 53.86% obtained using only LiDAR. Similar improvements are observed in AA and κ . For the Trento dataset, the fusion strategy yields an OA of 99.06%, a substantial increase over the 96.76% from HSI alone and 87.01% from LiDAR alone. Comparable trends are seen in the MUUFL dataset. Overall, these results demonstrate that fusing HSI and LiDAR data effectively leverages the strengths of both modalities,

resulting in superior classification performance across all datasets. This fusion approach is especially beneficial in complex scenarios, where it significantly enhances both accuracy and robustness.

The ablation study also highlights the contributions of the MMFB and CMFEB modules to the overall performance of the model on the Houston, Trento, and MUUFL datasets (see Table 5). When examining the MMFB independently, there is a noticeable improvement in performance, particularly on the Houston dataset, where the OA increases to 92.86%, the AA to 93.43%, and the κ to 91.87%. Similarly, the Trento and MUUFL datasets exhibit strong results with this module, underscoring its effectiveness in enhancing feature fusion across modalities. Likewise, the CMFEB also contributes positively when used alone, achieving an OA of 92.51% on the Houston dataset, 98.72% on the Trento dataset, and 90.83% on the MUUFL dataset, along with corresponding increases in AA and κ . This module proves especially beneficial for extracting complementary features across modalities, refining the model's capacity to distinguish between different classes. These findings clearly demonstrate that both modules are crucial to the overall performance, each offering unique advantages. When combined, the synergy between

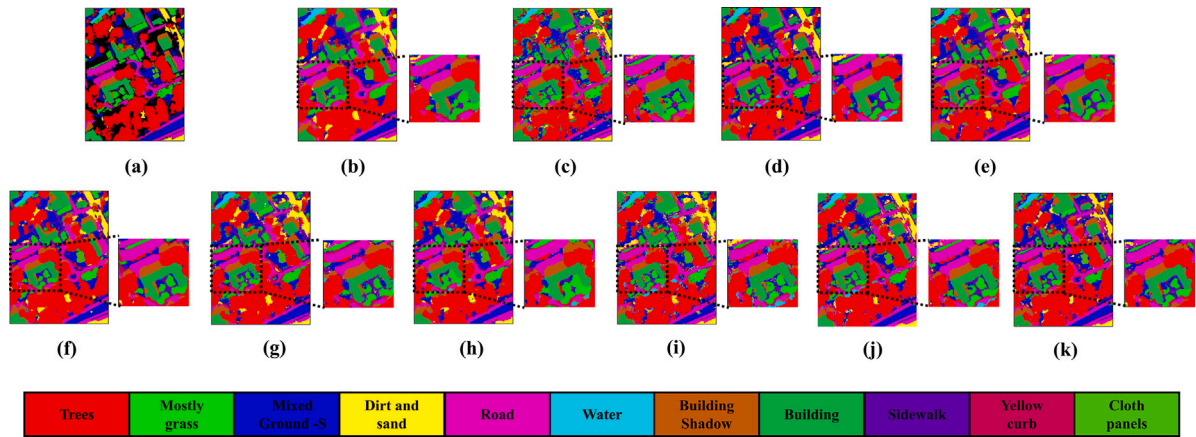


Fig. 11. The classification maps of different models on the MUUFL dataset. (a) ground truth. (b) CALC. (c) GAMF. (d) GLTNet. (e) M2FNet. (f) MS2CANet. (g) S²ENet. (h) MFT. (i) ExViT. (j) HLMamba. (k) Proposed.

feature fusion and cross-modality extraction results in the highest OA, AA, and κ values across the datasets. Thus, the interplay between these two modules is essential for maximizing model performances.

6. Conclusions

In this paper, we make a preliminary attempt to apply the Mamba architecture to the joint classification task of HSI and LiDAR data, achieving performance that surpasses state-of-the-art CNN-based and Transformer-based methods. Specifically, we first employ a dual-branch VSSB structure with shared weights to independently capture the distinctive features of HSI and LiDAR. The VSSB structure offers linear computational complexity while maintaining the ability to model global features. These processed features are then passed through the CMFEB module, where thorough interaction between the two modalities occurs via linear layers, DWConv and the CSSM structure. Finally, the feature embeddings are fed into the MMFB module, where the modality-specific features undergo secondary enhancement. An attention mechanism is introduced to select salient information from each modality, further enhancing the network's discriminative power. Extensive experiments demonstrated that the proposed model delivers satisfactory classification performance across all three datasets. In our future work, we plan to design more efficient Mamba modules to fully exploit the distinct characteristics of HSI and LiDAR. For instance, we will develop specialized Mamba modules for spectral feature extraction tailored to the complex spectral bands of HSI, and create dedicated modules for extracting elevation information from LiDAR data. Furthermore, we intend to extend Mamba's high performances to other HSI-related tasks, such as HSI change detection, target detection, and other downstream tasks.

CRedit authorship contribution statement

Yiyan Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Hongmin Gao:** Validation, Supervision, Funding acquisition. **Zhonghao Chen:** Formal analysis, Data curation, Conceptualization. **Chenkai Zhang:** Visualization, Validation, Software. **Pedram Ghamisi:** Validation, Supervision. **Bing Zhang:** Validation, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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